

Test results part 2

Behavioral Intent-Based Deception

Phase 2 Validation Report — End-to-End Deception Loop

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Lab: VMware — shadowtier-ips + shadowtierhoneynet + Kali Linux

1. Overview

This report documents the complete end-to-end validation of Phase 2 — the deception engagement loop. Phase 1 confirmed that all six behavioral detection rules fire correctly against attack traffic. Phase 2 validates the full chain: detection triggers automatic DNAT redirection, attacker traffic is silently routed into the deception environment, and the fake financial server engages the attacker with convincing false responses while logging all interactions.

The attacker believes they are interacting with a real financial network perimeter at 192.168.20.1. They are inside a controlled deception environment at 192.168.30.2. Every action they take is logged. Every credential they receive is false. Every success message they see is manufactured.

2. Phase 2 Architecture

2.1 Three-VM Lab Setup

- Kali Linux (192.168.20.2) — Attacker machine. Generates attack traffic targeting 192.168.20.1.
- shadowtier-ips (192.168.20.1 / 192.168.30.1) — The brain. Suricata detects behavioral patterns. Redirect engine fires DNAT rules. iptables routes classified traffic to deception network.
- shadowtierhoneynet (192.168.30.2) — The maze. Isolated deception VM running fake financial server. Logs all attacker interactions. Invisible to attacker — he never chose to go there.

2.2 Phase 2 Components Built

- fake_bank.py — Python deception server simulating a financial web application. Responds to /login, /auth, /admin, /api with convincing fake data including session tokens, admin credentials, and account balances.
 - shadowtier_redirect.py — Python trigger script watching Suricata fast.log in real time. Extracts attacker IP from Shadow Tier alerts and automatically injects iptables DNAT PREROUTING rules.
 - iptables DNAT PREROUTING — Silently rewrites destination of attacker traffic from 192.168.20.1 to 192.168.30.2 without attacker awareness.
 - iptables POSTROUTING MASQUERADE — Ensures reply traffic from deception VM routes correctly back through shadowtier-ips to attacker.
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3. Component 1 — Deception Server Initialization

The fake financial server was deployed on the isolated deception VM (shadowtierhoneynet) at /opt/deception-server/fake_bank.py. On startup it confirms the maze is active and begins listening on port 80 for incoming connections.

```
malak@shadowtierhoneynet:~$ sudo mkdir -p /opt/deception-server
[sudo: authenticate] Password:
malak@shadowtierhoneynet:~$ sudo cp /tmp/fake_bank.py /opt/deception-server/fake_bank.py
malak@shadowtierhoneynet:~$ sudo python3 /opt/deception-server/fake_bank.py
Shadow Tier Deception Server starting on port 80...
[2026-07-18 06:57:11] Deception server initialized - maze is active
```

Figure 1: `fake_bank.py` deployed and initialized on `shadowtierhoneynet`. "Shadow Tier Deception Server starting on port 80 — maze is active." Deception environment confirmed operational.

4. Component 2 — Redirect Engine

4.1 Engine Startup

The `shadowtier_redirect.py` script was deployed on `shadowtier-ips` and launched to watch Suricata alerts in real time. The engine tails the Suricata `fast.log` file and parses every line for SHADOWTIER-prefixed alerts.

```
fake_bank.py
malak@shadowtier-ips:~$ sudo python3 /tmp/shadowtier_redirect.py
[sudo: authenticate] Password:
[SHADOW TIER] Redirect engine started - watching for alerts...
-
```

Figure 2: Redirect engine started on `shadowtier-ips`. "[SHADOW TIER] Redirect engine started — watching for alerts..." Engine confirmed watching Suricata `fast.log`.

4.2 Automatic Attacker Detection and Redirection

When Kali launched attack traffic, the redirect engine parsed the Suricata alert, extracted the attacker IP (192.168.20.2), and automatically fired the iptables DNAT rule — routing all subsequent traffic from that IP into the deception environment without any manual intervention.

```
fake_bank.py
malak@shadowtier-ips:~$ sudo python3 /tmp/shadowtier_redirect.py
[sudo: authenticate] Password:
[SHADOW TIER] Redirect engine started - watching for alerts...
[SHADOW TIER] Attacker detected: 192.168.20.2 - redirecting to maze
[SHADOW TIER] 192.168.20.2 now inside the maze
[SHADOW TIER] Attacker detected: 192.168.20.1 - redirecting to maze
[SHADOW TIER] 192.168.20.1 now inside the maze
-
```

Figure 3: Redirect engine firing automatically. "[SHADOW TIER] Attacker detected: 192.168.20.2 — redirecting to maze" followed by "192.168.20.2 now inside the maze." Zero manual intervention required.

This confirms the automated pipeline: Suricata alert → script parses alert → iptables DNAT injected → attacker redirected. The time between detection and redirection is sub-second.


```
[2026-07-18 12:20:33] DECEPTION Server initialized - fake is active

[2026-07-18 12:36:03] ATTACKER 192.168.30.1 POST /login - DATA:
[2026-07-18 12:36:03] DECEPTION: Sent fake login success to 192.168.30.1
[2026-07-18 12:36:03] AT

TACKER 192.168.30.1 POST /login - DATA:
[2026-07-18 12:36:03] DECEPTION: Sent fake login success to 192.168.30.1
```

Figure 6: Deception VM attacker_log.txt showing first captured interaction. "ATTACKER 192.168.30.1 POST /login" followed by "DECEPTION: Sent fake login success to 192.168.30.1." Attacker interaction logged, fake response confirmed sent.

6.3 Full Attack Simulation — All Four Endpoints

A complete attack simulation was run from Kali targeting all four financial endpoints — /login, /admin, /api, and /auth — three times each. The attacker consistently received convincing fake responses for every request while inside the deception environment.

```
(kali@kali)-[~]
└─$ for i in {1..3}; do curl -X POST http://192.168.20.1/login; done
for i in {1..3}; do curl -X POST http://192.168.20.1/admin; done
for i in {1..3}; do curl -X POST http://192.168.20.1/api; done
for i in {1..3}; do curl -X POST http://192.168.20.1/auth; done
{"status": "success", "token": "eyJhbGciOiJIUzI1NiJ9.eyJpbnNpdGciOiJIUzI1NiJ9.fake.token", "message": "Welcome back. Session established.", "session_id": "SID-7842-ALPHA"}{"status": "success", "token": "eyJhbGciOiJIUzI1NiJ9.fake.token", "message": "Welcome back. Session established.", "session_id": "SID-7842-ALPHA"}{"status": "success", "token": "eyJhbGciOiJIUzI1NiJ9.fake.token", "message": "Welcome back. Session established.", "session_id": "SID-7842-ALPHA"}{"status": "access_granted", "panel": "admin_dashboard", "accounts": 47832, "total_balance": "$2,847,392,881.00", "message": "Admin panel loading.."}{"status": "access_granted", "panel": "admin_dashboard", "accounts": 47832, "total_balance": "$2,847,392,881.00", "message": "Admin panel loading..."}{"status": "access_granted", "panel": "admin_dashboard", "accounts": 47832, "total_balance": "$2,847,392,881.00", "message": "Admin panel loading..."}{"status": "ok", "api_version": "v2.3.1", "endpoints": ["/api/accounts", "/api/transfer", "/api/users"], "rate_limit": "1000/hour"}{"status": "ok", "api_version": "v2.3.1", "endpoints": ["/api/accounts", "/api/transfer", "/api/users"], "rate_limit": "1000/hour"}{"status": "ok", "api_version": "v2.3.1", "endpoints": ["/api/accounts", "/api/transfer", "/api/users"], "rate_limit": "1000/hour"}{"status": "authenticated", "role": "admin", "permissions": ["read", "write", "transfer"], "expires": "2026-12-31T23:59:59Z"}{"status": "authenticated", "role": "admin", "permissions": ["read", "write", "transfer"], "expires": "2026-12-31T23:59:59Z"}{"status": "authenticated", "role": "admin", "permissions": ["read", "write", "transfer"], "expires": "2026-12-31T23:59:59Z"}
```

Figure 7: Kali terminal showing full attack against all four endpoints. Attacker received: fake login token with session ID, fake admin panel access with \$2,847,392,881.00 account balance, fake API endpoint list, fake admin authentication with read/write/transfer permissions. All responses convincing, all data false.

6.4 Complete Deception Log — All Interactions Captured

```
[2026-07-18 12:40:12] DECEPTION: Sent fake login success to 192.168.30.1 [2026-07-18 12:40:12] AT
TACKER 192.168.30.1 POST /login - DATA:
[2026-07-18 12:40:12] DECEPTION: Sent fake login success to 192.168.30.1
[2026-07-18 12:40:13] ATTACKER 192.168.30.1 POST /admin - DATA: [2026-07-18 12:40:13] DECEPTION: Sent fake admin panel access to 192.168.30.1 [2026-07-18 12:40:1
3] ATTACKER 192.168.30.1 POST /admin - DATA:
[2026-07-18 12:40:13] DECEPTION: Sent fake admin panel access to 192.168.30.1
[2026-07-18 12:40:13] ATTACKER 192.168.30.1 POST /admin - DATA: [2026-07-18 12:40:13] DECEPTION: Sent fake admin panel access to 192.168.30.1 [2026-07-18 12:40:1
3] ATTACKER 192.168.30.1 POST /admin - DATA:
[2026-07-18 12:40:13] DECEPTION: Sent fake admin panel access to 192.168.30.1
[2026-07-18 12:40:13] ATTACKER 192.168.30.1 POST /admin - DATA: [2026-07-18 12:40:13] DECEPTION: Sent fake admin panel access to 192.168.30.1 [2026-07-18 12:40:1
3] ATTACKER 192.168.30.1 POST /admin - DATA:
[2026-07-18 12:40:13] DECEPTION: Sent fake admin panel access to 192.168.30.1
[2026-07-18 12:40:13] ATTACKER 192.168.30.1 POST /api - DATA: [2026-07-18 12:40:13] DECEPTION: Sent fake API response to 192.168.30.1 [2026-07-18 12:40:13] ATTAC
KER 192.168.30.1 POST /api - DATA:
[2026-07-18 12:40:13] DECEPTION: Sent fake API response to 192.168.30.1
[2026-07-18 12:40:13] ATTACKER 192.168.30.1 POST /api - DATA: [2026-07-18 12:40:13] DECEPTION: Sent fake API response to 192.168.30.1 [2026-07-18 12:40:13] ATTAC
KER 192.168.30.1 POST /api - DATA:
[2026-07-18 12:40:13] DECEPTION: Sent fake API response to 192.168.30.1
[2026-07-18 12:40:13] ATTACKER 192.168.30.1 POST /api - DATA: [2026-07-18 12:40:13] DECEPTION: Sent fake API response to 192.168.30.1 [2026-07-18 12:40:13] ATTAC
KER 192.168.30.1 POST /api - DATA:
[2026-07-18 12:40:13] DECEPTION: Sent fake API response to 192.168.30.1
[2026-07-18 12:40:13] ATTACKER 192.168.30.1 POST /auth - DATA: [2026-07-18 12:40:13] DECEPTION: Sent fake auth token with admin role to 192.168.30.1 [2026-07-18
12:40:13] ATTACKER 192.168.30.1 POST /auth - DATA:
[2026-07-18 12:40:13] DECEPTION: Sent fake auth token with admin role to 192.168.30.1
[2026-07-18 12:40:13] ATTACKER 192.168.30.1 POST /auth - DATA: [2026-07-18 12:40:13] DECEPTION: Sent fake auth token with admin role to 192.168.30.1 [2026-07-18
12:40:13] ATTACKER 192.168.30.1 POST /auth - DATA:
[2026-07-18 12:40:13] DECEPTION: Sent fake auth token with admin role to 192.168.30.1
[2026-07-18 12:40:13] ATTACKER 192.168.30.1 POST /auth - DATA: [2026-07-18 12:40:13] DECEPTION: Sent fake auth token with admin role to 192.168.30.1 [2026-07-18
12:40:13] ATTACKER 192.168.30.1 POST /auth - DATA:
[2026-07-18 12:40:13] DECEPTION: Sent fake auth token with admin role to 192.168.30.1
-
```

Figure 8: Complete attacker_log.txt showing all interactions captured by the deception server. Every POST request logged with timestamp and source IP. Every deception response confirmed sent. /login, /admin, /api, /auth all engaged. Intelligence collection confirmed operational.

7. Phase 2 Validation Results

Component	Function	Result
Suricata Detection	Behavioral classification of persistent adaptive attacker	✓ VALIDATED (Phase 1)
Redirect Engine	Auto-parse alerts, inject DNAT rules without manual intervention	✓ VALIDATED
iptables DNAT	Silent traffic rewrite — attacker unaware of redirection	✓ VALIDATED
MASQUERADE NAT	Reply routing from deception VM back to attacker	✓ VALIDATED
fake_bank.py /login	Fake session token + success message returned to attacker	✓ VALIDATED
fake_bank.py /admin	Fake admin panel with \$2.8B balance returned to attacker	✓ VALIDATED
fake_bank.py /api	Fake API endpoints and rate limits returned to attacker	✓ VALIDATED
fake_bank.py /auth	Fake admin role with read/write/transfer permissions	✓ VALIDATED
Intelligence Collection	All attacker interactions logged with timestamp and source IP	✓ VALIDATED

End-to-End Loop: COMPLETE

Attacker Awareness of Redirection: ZERO

Intelligence Collected: ALL interactions logged with timestamp and source IP

Deception Believability: Attacker received full session tokens, admin access, and account data

8. The Complete Shadow Tier Loop — Proven

Phase 1 + Phase 2 together prove the complete architectural concept described in the executive summary:

Step 1 — Attacker probes 192.168.20.1 (real perimeter)

Step 2 — Layer Zero deception corrupts reconnaissance results (port 22 appears closed, 8080/3306 filtered)

Step 3 — Suricata classifies traffic: persistence + SYN scan + directed intent all detected simultaneously

Step 4 — Redirect engine fires DNAT rule automatically — sub-second response

Step 5 — All attacker traffic silently routed to 192.168.30.2 — attacker unaware

Step 6 — Deception server responds with fake credentials, fake admin access, fake financial data

Step 7 — Attacker believes he has compromised a real financial system

Step 8 — Every interaction logged: source IP, endpoint targeted, timestamp, response sent

This is the complete proof of the Shadow Tier concept. The attacker who refused to be stopped by dropping his packets has been guided into a controlled environment where every move he makes produces intelligence for the defender — while he believes he is advancing toward a real target.

9. Conclusion

Phase 2 validation is complete. The full end-to-end deception loop is proven in a live lab environment. Detection, classification, automatic redirection, deception engagement, and intelligence collection all function as designed.

The gap identified in the research paper — that no commercial DDoS mitigation platform offers a redirect-to-deception action alongside drop and rate-limit — has been addressed with a working implementation that demonstrates the concept is technically viable, not merely theoretical.

Combined with Phase 1 validation (6/6 detection rules, 100% detection rate), Shadow Tier now has a complete proof-of-concept ready for academic publication and institutional review.